FEB12001X - Resit 10 July 2024

FEB12001X Applied Microeconomics 23/24 (ESE) · 7 exercises · 52.0 points

1 Question 1

10.0 points · 5 questions

Consider the market for good X in country C. Demand for good X can be given by $P = 500 - \frac{2}{5}Q$. Supply of good X is given by $P = 100 + \frac{1}{10}Q$. Production of good X takes place in the country, but the production facilities are part of multinationals that are not owned by citizens of country C. Production leads to negative external effects, incurred by citizens of country C. Marginal external cost are given by $MEC = \frac{1}{10}Q$. The government of country C maximises a social welfare function that depends on the utility of the citizens of country C, but does not include the utility of the owners of the multinationals that produce good X. The government proposes to implement a unit tax of 100 euro on the production of good X.

Text

a Calculate the elasticities of supply and demand in the market equilibrium without tax on good X.

2.0 points · Open · 2/5 Page

Model answer

Equilibrium without tax $500 - \frac{2}{5}Q = 100 + \frac{1}{10}Q$ yields $Q = 800$ and $P = 180$. Elasticity of demand $|\varepsilon_d| = \frac{5}{2} \cdot \frac{180}{800} = \frac{9}{16}$. Elasticity of supply $\varepsilon_s = 10 \cdot \frac{180}{800} = \frac{9}{4}$.

+2 points

Both correct.

+1 point

One elasticity correct, or correct calculation of elasticities with mistake in market equilibrium.

b Derive the share of the tax burden borne by consumers if the government imposes this unit tax.

2.0 points · Open · 3/5 Page

Model answer

Equilibrium with tax $500 - \frac{2}{5}Q = 100 + \frac{1}{10}Q + 100$ yields $Q = 600$ and $P_C = 260$ and $P_P = 160$. Hence, consumers bear share $\frac{4}{5}$. Alternative: consumer share equals $\varepsilon_S/(\varepsilon_D + \varepsilon_S) = \frac{9}{4}/\left(\frac{9}{16} + \frac{9}{4}\right) = \frac{4}{5}$.

+1 point

Equilibrium with tax correct.

+1 point

Correct calculation of distribution of tax burden.

c How would the answer to question 1b) change if the demand would be less elastic? Explain why.

2.0 points · Open · 7/20 Page

Model answer

Less elastic demand implies that consumers have worse alternatives to consuming good X. This implies that the reduction in consumption following a price increase is smaller, such that producers can shift a larger part of the tax burden to consumers.

+2 points

Correct argumentation

There are N identical citizens living in country C. A citizen complains that the tax would hurt citizens more than that it helps them. A local economist argues that if the government would give each citizen an amount of money equal to fraction $\frac{1}{N}$ of the total tax revenue, citizens would be better off with the tax than without the tax.

Text

d Show that the economist is correct.

2.0 points · Open · 4/5 Page

Model answer

Note that citizens receive consumer surplus and incur external cost. Producer surplus is not received by them. Reduction in consumer surplus: before tax this is $\frac{1}{2}800(500-180) = 128000$. With tax, this is $\frac{1}{2}600\,(500-260) = 72000$. So CS goes down by 58000 euro. External cost reduce from $\frac{1}{2}80 * 800 = 32000$ to $\frac{1}{2}60 * 600 = 18000$, so by 14000. The government receives 100*600=60000 euro in tax revenue. Hence, if this tax revenue is returned to citizens, the citizens are better off with
the tax than without.

+2 points

Correct calculation

+1 point

Incorrect use of producer surplus, or incorrectly not including CS, EK, or tax revenue, but otherwise correct approach.

e In this exercise, the level of a Pigouvian tax on good X is equal to 66,67 euro per unit. Give a normative explanation for why the government in this exercise would implement a tax that is higher than the Pigouvian level.

2.0 points · Open · 2/5 Page

Model answer

Multiple answers possible. Main answer: by implementing a tax above the Pigouvian level, the government can increase the utility of citizens, as it can `capture' part of the producer surplus that is otherwise obtained by the owners of the multinationals. As these are not included in the government's social welfare function, doing so would increase social welfare (at the expense of the owners of the multinationals).
Alternative: By generating more tax revenue via this tax, the government may be able to reduce other distortionary taxes.
Note: given that all citizens are identical, redistributional arguments are not valid in this case.

+2 points
Valid normative explanation

+1 point
Redistributional argument

0 points
Positive explanation

2 Question 2

6.0 points · 3 questions

Suppose there are two individuals, Anne and Ben, who both derive utility from the consumption of a public good. Anne's valuation of the good in euro is given by

$$U_A = a\ln(q_A)$$

where q_A is the quantity consumed by Anne and $\ln(\cdot)$ is the natural logaritm. Ben's valuation in euro is given by

$$U_B = b\ln(q_B)$$

where q_B is the quantity consumed by Ben. Supply of the public good on the market is perfectly elastic. Marginal production cost are equal to the price $p = MC = c$ euro. We assume that $a > b > c > 0$.

Text

a Derive the socially efficient level of consumption of Anne and Ben.

2.0 points · Open · 2/5 Page

Model answer

As it is a public good, it must be that $q_A = q_B = q$. $MU_A + MU_B = MC$ gives $\frac{a}{q} + \frac{b}{q} = c$ which yields $q = \frac{a+b}{c}$.

+2 points

Correct

+1 point

Correct approach with incorrect marginal utilities

Suppose that the public good is provided by the market. Anne and Ben decide independently how many units of the public good to buy (of course taking the presence of the other into account).

Text

b Determine how many units of the public good Ben consumes in the Nash-equilibrium that arises on the market.

2.0 points · Open · 3/5 Page

Model answer

In the Nash equilibrium Anne provides $\frac{a}{c}$ units and Ben 0, such that Ben consumes $\frac{a}{c}$ units. If Ben provides 0, it is optimal for Anne to provide such that $MU_A = c \rightarrow \frac{a}{q_A} = c$ so that $q_A = \frac{a}{c}$ units. Then $MU_B = \frac{b}{a/c} < c$ (as $a > b$), which implies that Ben optimally provides 0. This is the only Nash equilibrium.

+2 points

Correct

+1 point

Correct Nash equilibrium, missing or wrong conclusion ("Ben consumes 0").

Now suppose the government offers a subsidy per unit of production to producers of the public good. The Pigouvian level of the subsidy equals $\frac{bc}{a+b}$ euro. This expression shows that the Pigouvian subsidy increases in b.

Text

c Explain why the Pigouvian subsidy increases in b. (note: give an economic explanation, not `math in words')

2.0 points · Open · 3/10 Page

Model answer

Given that Anne provides in the Nash equilibrium, the marginal utility of Ben is the marginal external benefit. The MEB increase in b. Hence, to make Anne internalise Ben's benefits, the Pigouvian subsidy needs to increase in b as well.

+2 points

correct

0 points

math in words

3 Question 3

10.0 points · 4 questions

Peter derives utility from the consumption of his disposable income C, as given by his utility function

$$U = \sqrt{C}$$

When Peter does not fall ill, his disposable income is 2500 euro. If he does fall ill, he has to pay 1600 euro health care cost, implying that his disposable income is only 900 euro. The probability that Peter falls ill depends on his lifestyle. If he chooses a healthy lifestyle, the probability that he falls ill is 10 percent. If he chooses an unhealthy lifestyle, the probability that he falls ill is 30 percent. A healthy lifestyle leads to a loss in utility equal to c, where $0 < c < 5$. Peter maximises his expected utility.

Text

a Determine for which levels of c Peter chooses a healthy lifestyle.

2.0 points · Open · 2/5 Page

Model answer

Unhealthy lifestyle yields $EU = 0.7\sqrt{2500} + 0.3\sqrt{900} = 44$. Healthy lifestyle yields $EU = 0.9\sqrt{2500} + 0.1\sqrt{900} - c = 48 - c$. Hence, Peter chooses healthy if $c < 4$.

+2 points

correct

0 points

Utility of expected income rather than expected utility

Peter lives in a country where people can decide for themselves whether to insure against health care cost. Insurance companies cannot observe whether people have a healthy lifestyle or an unhealthy lifestyle. The insurance premium is equal to the expected payout as expected by the insurance company (`actuarially fair').

Text

b Explain why an insurance that pays out 1600 euro to Peter when he falls ill has an insurance premium of 480 euro.

2.0 points · Open · 1/2 Page

Model answer

With full insurance, a healthy lifestyle only brings a utility loss of c, no benefits for Peter. With insurance he will choose for an unhealthy lifestyle. Anticipating this, the insurance company asks a premium of 0.3*1600 = 480 euro. Beyond the question: utility with insurance is $\sqrt{2500-480} = 44.94$, which is larger than utility without insurance.

+1 point

Choice Peter correctly discussed.

+1 point

Anticipation insurance companies correctly discussed.

Mary lives in the same country as Peter. She is identical to Peter in all respects (same utility function, same disposable income, same probabilities of falling ill and heath care cost, and the same loss of utility due to a healthy lifestyle), with one exception: When Mary falls ill, this not only leads to health care cost but also leads to a loss in her utility equal to 30. As always, we assume that this is known by insurance companies.

Fraction x of the citizens in this country is identical to Peter and the remaining fraction $1-x$ is identical to Mary. Assume that insurance companies are obliged to offer all citizens the same insurance premium.

Text

c Consider an insurance that pays out 1600 euro to citizens who fall ill. Calculate for which levels of x adverse election arises in the market for health insurance. (4 points).

4.0 points · Open · 1 Page

Model answer

All `Peters' live unhealthy, independent of the premium. All `Mary's' live healthy instead: $\sqrt{2500-p} - c - 0.1 * 30 > \sqrt{2500-p} - 0.3 * 30$ for any c between 0 and 5. If everyone would insure, the expected payout equals $x480 + (1-x)\,160$. This would be the premium p. Mary's are not willing to buy this insurance when $\sqrt{2500-p} - c - 0.1 * 30 < 0.9\sqrt{2500} + 0.1\sqrt{900} - c - 0.1 * 30$ such that $\sqrt{2500-p} < 48$ which yields $p > 196$. Hence, the Mary's only insure themselves when $x480 + (1-x)\,160 \leq 196$ which yields $x \leq \frac{9}{80} = 0.1125$. This implies that for any $x > \frac{9}{80}$, the premium is too high for Mary's to buy insurance, which implies that adverse selection arises.

+1 point

Mary's always choose healthy lifestyle

+1 point

Correct premium as function of x.

+2 points

Correct determination of minimum value of x via Mary's choice.

d Discuss one possible action that private parties on the market for health insurance in this exercise can take to mitigate the loss in efficiency due to the market failures.

2.0 points · Open · 9/20 Page

Model answer

Multiple correct answers. Most likely is a deductible / co-payment. This could reduce the moral hazard by Peters (making them choose healthy lifestyle even with insurance), which in turn would reduce the premium and avoid adverse selection. Alternative (less likely) is monitoring and making the insurance premium dependent on lifestyle. Screening is also a possibility, by offering one package with limited insurane and a low premium and one package with full insurance and a high premium, the efficiency loss due to adverse selection (not moral hazard!) could be reduced.

+2 points

Correct answer with valid argumentation

0 points

Answer with incorrect or no argumentation

4 Question 4

8.0 points · 3 questions

A not-for-profit organization employs one worker. When working at this organization, the utility of the worker is: $U = w + \gamma \ln(e) - \frac{1}{2}\theta e^2$, where U is utility, w is the salary, $e > 0$ is effort, and γ and θ are strictly positive. The salary w is independent of effort of the worker. It is sufficiently high so that the worker's participation constraint is satisfied. The first part of the utility function (w) represents the utility from private consumption. The logarithmic term represents the utility the worker derives from making a contribution to society, which he can do by exerting effort on the job. The last term represents the worker's cost of effort.

Text

a Derive the level of effort that maximizes the worker's utility.

2.0 points · Open · 1/2 Page

+1 point

Take the derivative of the utility function and set equal to zero: $\gamma/e - \theta e = 0$

+1 point

Solve for e gives $e = \sqrt{\frac{\gamma}{\theta}}$

b Assume the worker obtains utility equal to zero when not working at this organization. Also assume that the worker chooses to work for this organization, unless he is better off by not doing that. Show that the minimum level of the wage the organization needs to offer to induce the worker to work at this organization equals: $-\gamma \ln\left(\sqrt{\frac{\gamma}{\theta}}\right) + \frac{1}{2}\theta\left(\sqrt{\frac{\gamma}{\theta}}\right)^2$.

2.0 points · Open · 1/2 Page

+1 point

The worker is willing to work at this organization when it yields a utility that is at least equal to zero. Set Utility U equal to zero.

+1 point

Solving for the wage yields the expression in the question.

c Let's denote the minimum level of the wage that the organization needs to offer to induce the worker to work at this organization by $w^{\min}$. As is clear from the formula in question 4b, $w^{\min}$ depends on the parameter γ in an interesting way. Explain why this is the case. Start with one sentence describing the meaning of the parameter γ. Next describe in three clear sentences, and using the formula in question 4b, three reasons why the parameter γ affects $w^{\min}$. Give the economic interpretation, so do not use "math in words".

4.0 points · Open · 3/4 Page

+1 point

The parameter γ measures how much the worker cares about making a contribution to society.

+1 point

The more the worker cares about this, the more the worker enjoys any given level of effort.

+1 point

Moreover, the worker works more, which further raises his utility from making a contribution to society.

+1 point

Lastly, since the worker works more, the worker has a higher cost of effort.

5 Question 5

8.0 points · 4 questions

The CEO of a large retail chain wants to find out whether local store managers discriminate minorities when selecting job applicants for a job interview. The CEO wants to obtain reliable evidence on this, not just an indication.

Text

a Propose a research design that will deliver reliable evidence on whether local store managers discriminate minorities or not. Use at most three sentences.

2.0 points · Open · 1/2 Page

+1 point

Send to the stores with job openings a large number of fake job appplications, of which some clearly come from a minority group and others from the majority group.

+1 point

Compare the response rate to applications from a minority group with the response rate to applications from the majority group to see whether minority status systematically affects the chances of a job interview.

Suppose the research showed that there is discrimination against minorities. The CEO now wants to find out whether there is taste-based discrimination or statistical discrimination.

Text

b Propose a research design that will provide insight into whether there is taste-based or statistical discrimination or both.

2.0 points · Open · 1/2 Page

+1 point

Send a large number of fake job applications from majority and minority job seekers and now also include additional relevant information to some of the applications but not to others (randomly chosen).

+1 point

If the difference in response rates between majority and minority job seekers diminishes in response to additional information, the discrimination is at least to some extent statistical discrimination.

c Propose one way in which taste-based discrimination when selecting job applicants for job interviews can be eliminated in practice. Also provide a short description how this works and why it eliminates taste-based discrimination, in a few clear sentences.

2.0 points · Open · 2/5 Page

+2 points

Blind all aspects in job applications that should not play a role in the selection for interviews. For instance, anything that refers to the age or gender of job applicants could be removed (or made invisible) before the recruiters have a look at the applications and make a choice whom to invite for an interview.

The CEO is not convinced by your idea and decides to follow an alternative approach. He sends all store managers to a diversity training. One year later, the personnel data of the company show that a larger share of new hires has a minority background than in the year before the training.

Text

d Did the diversity training cause this increase? Give a short explanation.

2.0 points · Open · 3/10 Page

Model answer

We cannot be sure. The reason is that we do not know what would have happened when the store managers had not followed the training.

+2 points

6 Question 6

10.0 points · 4 questions

A company thinks about implementing team production and team incentives. At this moment, workers earn a base salary equal to a and a bonus based on individual production $b > 0$ per unit of production. The production of worker i is described by $q_i = ve_i$ where q_i is the production of worker i, v the ability, and e_i the level of effort by worker i. Workers have identical preferences. The utility function is: $U_i = w_i - \gamma\left(w_j - w_i\right) - \frac{1}{2}\theta e_i^2$, where U_i is the utility of worker i, w_i the total salary of worker i, w_j the total salary of worker j, and $0 < \gamma < 1$ and $\theta > 0$ are constants. Each worker chooses his effort independently from the other so as to maximize his own utility.

Text

a Describe in one clear sentence a possible meaning of the term $-\gamma\left(w_j - w_i\right)$ in the utility function of worker i. Give an economic interpretation, so no 'math in words'.

2.0 points · Open · 1/2 Page

+1 point

The worker cares about how his wage compares to the wage of the other worker.

+1 point

This may arise from status concerns, envy, pride, or a combination thereof. (You need to mention just one of them.)

b What is the level of production that worker i achieves under the current remuneration system?

2.0 points · Open · 1/2 Page

+1 point

Write down the utility function and substitute $a + bve_i$ for the wage w_i. Take the derivative to e_i and set equal to zero.

+1 point

Solve for e_i which gives $e_i = \frac{(1+\gamma)bv}{\theta}$. Substitute into the production function and you get $q_i = \frac{(1+\gamma)bv^2}{\theta}$

c Examine whether the parameter γ influences the level of effort that worker i chooses under the current remuneration system and give a brief explanation.

2.0 points · Open · 1/2 Page

+1 point

From the result it is clear that effort increases with γ. (If you made a calculation error, you get full point for a conclusion that is consistent with your erroneous calculation)

+1 point

The worker enjoys earning more than the other worker, the way to try and achieve that is by working more.

The company decides that the workers must work in a team of two. Each worker earns the same base salary, a, and a bonus on the basis of the team production equal to $\frac{1}{2}b$ per unit of production, where b equals the level of the bonus under individual production. Under team production the production of the team is equal to: $\kappa v\left(e_i+e_j\right)$ where κ is larger than or equal to one. (As usual, we assume that employees cannot contract with each other and do not make any income transfers to each other. We also assume that a is sufficiently high so that both workers want to continue working for this company.)

Text

d Is production higher or lower after the company has implemented team production and team rewards? Determine the condition.

4.0 points · Open · 7/10 Page

+1 point

Write down the utility function and substitute the new production function and the new wage scheme.

+1 point

Differentiate to e_i set equal to zero and solve which yields: $e_i=\frac{b\kappa v}{2\theta}$. (As a side note, note that γ plays no role anymore because when a worker increases effort, this raises both workers' wages with the same amount.)

+1 point

Substitute into the production function. Recognize that worker j will choose the same effort level. And so substitute the same effort for e_j. Team output is $\frac{b\kappa^2v^2}{\theta}$.

+1 point

Compare with the joint individual output. The resulting condition for higher production is $\kappa>\sqrt{2\left(1+\gamma\right)}$

7 Additional answer space

0.0 points · 5 questions

Only use this extra space if you made a mistake in one of your answers. Please do not forget to **indicate in the answer box of that particular question where the correct answer can be found**. Otherwise, the answers in the additional space below will not be reviewed.
NB: Do **not** use this additional space for answers that are too long.

Text

a Additional answer box 1

Open · 4/5 Page

b Additional answer box 2

Open · 1/2 Page

c Additional answer box 3

Open · 1/2 Page

d Additional answer box 4

Open · 1/2 Page

e Additional answer box 5

Open · 1/2 Page